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A Generative AI Framework for Automatic Comic Strip Generation

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Abstract—Research over years has proved that the human brain can remember information better when it is seen compared to when it is heard or read. This highlights the power of visual over other senses. Based on this principle this research aims to deliver a comic generator using artificial intelligence to create images and comic strip based on input text. The model integrates natural language processing technique to understand input, extract character and understand storyline. The generative models focus on generating image along with suitable comic characters along with proper emotions as well as a textual generative model for generating comic strips and dialogue bubble. This project has a capability to develop 3 comic strips to 6 comic strips based on users choice. Once comic strips are generated a layout engine is utilized to arrange dialogue bubble on images. This project aims to reduce the artistic effort of and deliver comic content. This product can be used in various domains including writers, educators, students and content designer to make storytelling more effective and rapid.

Index Terms—Artificial intelligence, Natural language processing, Generative model, Layout engine.

I. INTRODUCTION

Comics combine the traditional art of story telling with creativity, making it more engaging and emotionally impactful. They are utilized for various purposes other than entertainment such as education, productivity and even as stress bursters. Comics are enjoyed by people of all age groups. Reading comics improves a person's creative thinking and induces interest in reading which strengthens the brain and improves language skills.

In this modern computerised world reading story books has decreased to a large extent and the availability and accessibility of comics has reduced significantly. Printing of comics is not cost efficient and once printed the format and style or language cannot be altered. Our model Generates comic strips from the scenarios and dialogues given as input. The strip consists of images of characters in the background of the given scenario along with descriptive text and dialogues of the characters. The style and font of the comic generated can be modified according to the context. This model generates images in panels of each scenario. It makes use of LLM to generate the descriptive text for each panel which is then validated using the backend JSON dataformat. The

image for each panel ids generated using stability AI which is a efficient platform for generative models. Once all the panels are generated for the given scenario with necessary images and text, the model combines it into a comic strip consisting all the panels in the required order. Once the strip is generated the user can download this comic for further usage.

This increases accessibility, time efficiency and cost efficiency to a significant extent. It reduces the time needed for printing the comic books and is also cost efficient. These comics can also be used to deliver useful insights on various fields such as education, morality, ethics, history, spirituality which nurtures the development of noble behaviour in an individual. Knowledge can be delivered in a fun and interactive manner using comic generators.

II. LITERATURE REVIEW

The accelerated innovations in area of Artificial Intelligence has brought in new scope for projects related to generative AI. Combining techniques such as Natural language processing for processing the text and CNN for processing the image part has proven effective in enriching the model's capability to analyze and interpret the emotional, spatial and visual context of the given description. This analysis helps the stable diffusion model and DALL-E model in converting the text into visuals. Evaluation metrics such as accuracy, f1 Score, precession, and recall help in determining how relevant the generated image is to the given text[1].

The generation of comic strips can be enhanced by extracting semantic information from many datatypes, that is video , audio and text. A model was developed to extract video frames from a clip and group them into frames with dialogue and frames without dialogue using GIST similarity. The emotional information was derived from analysing the subtitles and audio sentiment which was used for generating the text balloons by feeding them into a SVM for selecting the shape and style of the balloon based on the detected emotion. The output panels were generated in manga like style using difference of Gaussian algorithm. These panels

were then arranged into the given order using a data-driven stylistic layout model[2].

Studies tell that people who have the habit of reading comics are more empathetic towards the societal issues. A web application is more accessible to users than printed comics. This application generates comic strips based on the text entered by the user by classifying the text using fast text techniques and support vector machines and generating visuals using methods like generation of text bubbles, character recognition and voiceover interpretation[3]. The deployment of sequential models contribute significantly for generation of comic strips. Based on the narration of the story which is considered as the criteria for the formation of comic strips can be performed using AI models. The layers of these models make use of decision making capabilities to transform theories into imageries making it more interactive[4].

A comparative study performed by comparing 15 AI comic strip generating platforms showed that integration of generative AI in generating comics results in efficient, faster and cost effective process where creativity meets automation. This study also showed a comparative analyses of various pricing of these platforms from freemium to paid subscription which concluded that free platforms attract greater number of users especially beginners[5]. Another study performed on how reading comics benefits education showed that reading comics induces creative and artistic thinking in students. Diverse information which can be complex to explain can be effectively delivered to students by visual representations with descriptive texts as presented in a comic[6].

III. METHODOLOGY

The research aims to design a product that is capable of understanding human language and generate suitable storyline according to user requirement. Based on generated storyline, the product also generates comic characters with dialogue bubble and aligns all generated panel in order. The system understand human input through natural language processing and uses generative text and image models to generate required comic panels.

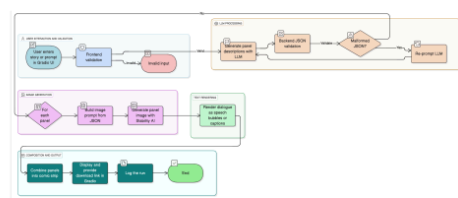


Fig. 1. Workflow

The pipeline begin with a gradio application that serve as front-end where user enters their input in text format. Gradio being a lightweight application it is easy to deploy in any platform. This input can be a situation or description of story background. It may or may not include characters. The input is validated before being submitted to model for generation of output. The users input is later engineered using appropriate prompt and then submitted to model. The product

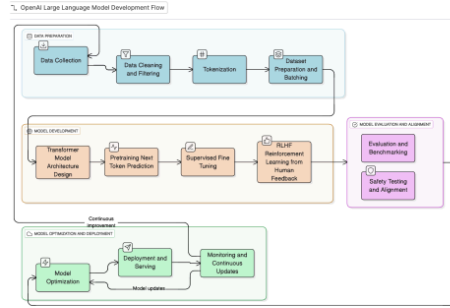


Fig. 2. OpenAI LLM architecture

contains two models to perform task. First model in pipeline is OpenAI Large Language Model (LLM). This model is capable of understanding human input, extract character based information from text, read emotions suggested in input and other environmental details explained in prompt. All the above details are achieved using natural language processing techniques like embeddings, lemmatization, sentimental analysis and name entity recognition.

Using JSON format text generative model communicates with image generative model. In this JSON, number of panels, characterization and scene description is sent. Once OpenAI LLM understand scene its interpretation is passed onto validation layer to cross check JSON format is according to desired requirements. This is one of the error-handling strategies to prevent execution failures.

Validation layer sends image request to Stability AI, this is capable of visual generation. This can generate images in comic style. This model is pre-trained with various images in comic style and is capable of generating any image in comic strips. After all images are generated, the dialogue and image panels are placed at right location using PIL. PIL, Pillow is a python visual tool that can arrange images in sequence format. This act as layout engine.

Finally, generated individual panel is arranged in sequence order. The user can view and download panels. Thus the project combines Large Language Model, intelligent prompt-

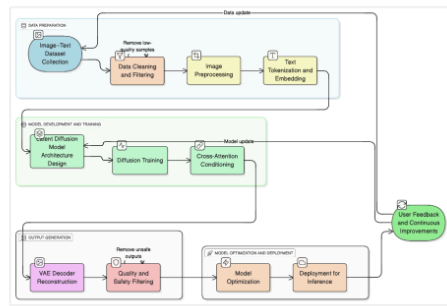


Fig. 3. Stability AI architecture

ing, diffusion models and simple user friendly interface to generate comic images based on storyline.

IV. RESULTS AND DISCUSSION

A. Results

AI comic generator can transform text to conceptual images in comic style. This process in completely automated using Machine learning operations (Mlops) pipeline. Using multi-modal for various operation such as text generation and context understanding another model for image generation and layout engine, required outcome was obtained. The performance of this product was observed that it was operating smoothly on CPU based environment.

Despite the product contains two different API's for different models it requires 10 to 13 seconds on an average to generate a panel of 3 images and 18 to 22 seconds to generate a panel of 6 images. The below table gives a descriptive analysis of number of panels and average time take to generate panels in seconds.

TABLE I
TABLE OF OUTCOME

Number of Panels	Average Time (seconds)
3 Panel	10–13 seconds
4 Panels	12–16 seconds
5 Panels	14–19 seconds
6 Panels	18–22 seconds

The visual quality of image remain constant in every generation. Usage of validation layer improves error handling. Gradio interface provides simple and easy access to user and speed up the process. Overall, the result conforms relevant, multi-row comic layout with dialogue bubbles is generated.

B. Discussion

This research enforces utilization of generative artificial intelligence model in effective storytelling. This module can be used by educators to enhance explainability of theoretical

topic in easy manner. Students can use this model to analyze and re-imagine concepts. Writers can spread their writings to large audience. Below image showcase an example input along with comic strip output.



Fig. 4. Output when input was explain barter system in India style.



Fig. 5. Output when input was explain barter system in American style.

V. FUTURE WORK

Currently our model generates 6 panels to form a comic strip. This can be improvised by integrating AI models to learn and interpret the number of panels required according to the given story. A web or a mobile application can be developed where the interactive user API can prompt the user to enter the scenario of the story and the comic generated can be downloaded in the form of documents.

Audio narration along with text description for the comic images can be integrated to overcome readability issues. The user can be promoted to input the language in which the comic text has to generated. This can be achieved using translators. Also the style and format of the comic can be customized according to the user inputs. These models show great scope in improving many sectors such as delivering quality education and knowledge about diverse cultures and spreading social awareness.

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